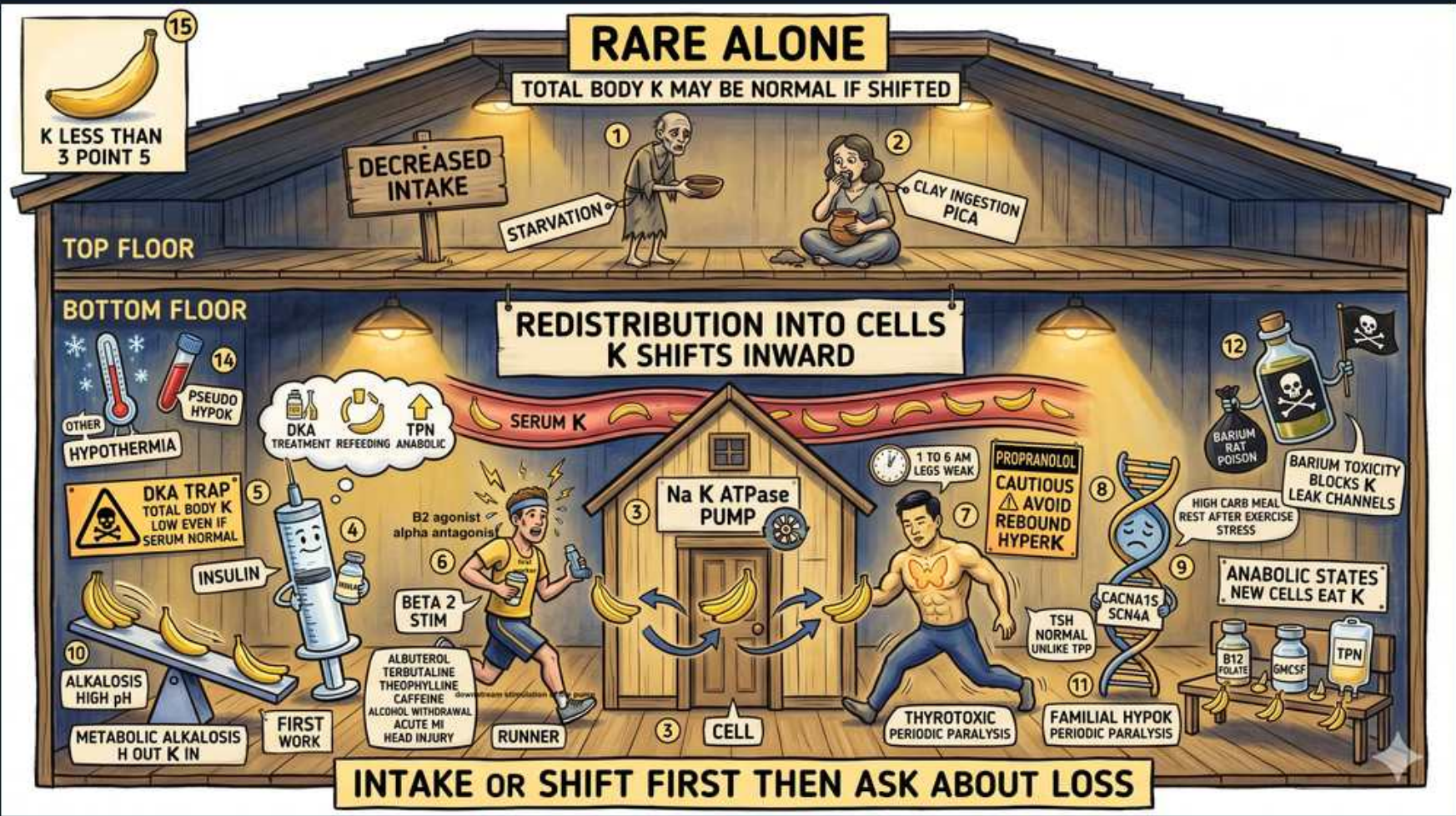


Hypokalemia — redistribution / transcellular shift



1. Shift = total body K normal

WHAT YOU SEE

Banner: RARE ALONE — total body K may be normal if shifted

WHAT IT ENCODES

Redistribution moves K INTO cells without changing total body K. Serum K is low but there is no true deficit, so the kidney shows low urine K and the patient is normotensive. This branch must be excluded before chasing a renal or GI leak.

BOARD TRAP

Aggressively replacing K in a pure shift (e.g. periodic paralysis, beta-agonist overdose). When the trigger resolves, K floods back out of cells → dangerous REBOUND hyperkalemia. Give modest doses and monitor.

2. K shifts inward

WHAT YOU SEE

Mid banner: redistribution into cells, K shifts inward; serum K bar

WHAT IT ENCODES

Mechanism: anything that stimulates the Na/K-ATPase (insulin, beta-2) or alters transmembrane charge (alkalosis) pulls K into cells → serum K falls. Each is reversible once the driver is removed, distinguishing shift from true loss.

BOARD TRAP

Assuming a low serum K always means a body deficit. In shift states total body K is normal; the giveaway is a rapid, large fall in K (or rapid normalization) tied to a metabolic trigger rather than ongoing losses.

3. Na/K ATPase pump

WHAT YOU SEE

Central glowing Na K ATPase pump with bananas (K) moving in

WHAT IT ENCODES

The Na/K-ATPase is the engine of redistribution: it pumps 3 Na out / 2 K in. Insulin and beta-2 agonists upregulate it, driving K intracellularly — the same mechanism exploited to TREAT hyperkalemia (insulin+glucose, nebulized albuterol).

BOARD TRAP

Forgetting that the hyperkalemia-lowering drugs (insulin, albuterol, bicarbonate) cause hypokalemia by the identical shift mechanism — over-treating a hyperkalemic patient flips them hypokalemic.

4. Alkalosis → K in

WHAT YOU SEE

Banner left: alkalosis high pH, metabolic alkalosis = H out K in, first work

WHAT IT ENCODES

Alkalemia: H⁺ leaves cells to buffer the high pH, and K⁺ enters to maintain electroneutrality → hypokalemia. Roughly serum K falls ~0.3 mmol/L per 0.1 rise in pH. Conversely acidosis shifts K out (raising serum K).

BOARD TRAP

Aggressively replacing K based on the shift alone in an alkalotic patient — correcting the alkalosis shifts K back out and can overshoot. Treat the underlying alkalosis; replace K cautiously and reassess.

5. Insulin / DKA treatment

WHAT YOU SEE

Insulin syringe; DKA treatment refeeding anabolic sign

WHAT IT ENCODES

Insulin drives K into cells via the Na/K-ATPase. Treating DKA or HHS — and refeeding/anabolic states — causes rapid intracellular K shift, so K must be monitored hourly and added to fluids when serum K <5.2 mmol/L.

BOARD TRAP

DKA trap: total body K is LOW even when the presenting serum K is normal or high (acidosis + insulin deficiency held it outside cells). Start insulin without K and you precipitate life-threatening hypokalemia — hold insulin if K <3.3.

6. DKA trap

WHAT YOU SEE

DKA TRAP warning: total body K low when serum normal

WHAT IT ENCODES

In DKA, insulinopenia and acidosis push K out of cells while osmotic diuresis dumps it in urine, so whole-body K is depleted despite a normal/high serum value. Insulin therapy unmasks profound hypokalemia within hours.

BOARD TRAP

Reading the initial serum K as reassuring and giving insulin alone. The exam answer: if K <3.3, HOLD insulin and replace K first; if 3.3–5.2 give insulin WITH K in the fluids; only if >5.2 give insulin without added K.

7. Beta-2 stimulation / runner

WHAT YOU SEE

Beta 2 stim, runner figure; albuterol, terbutaline, catecholamines, acute MI/head injury

WHAT IT ENCODES

Beta-2 agonists (albuterol, terbutaline, ritodrine) stimulate the Na/K-ATPase → K shifts into cells. Endogenous catecholamine surges do the same: acute MI, head injury, severe stress, plus theophylline and caffeine all drive K intracellularly.

BOARD TRAP

Missing iatrogenic cause: an asthmatic or COPD patient on continuous high-dose nebulized albuterol can become hypokalemic — it is a shift, not a renal or GI loss, so the fix is to reduce the beta-agonist, not hunt for a leak.

8. Thyrotoxic periodic paralysis

WHAT YOU SEE

Muscular figure; thyrotoxic periodic paralysis, TSH normal use this

WHAT IT ENCODES

Thyrotoxic periodic paralysis: excess thyroid hormone amplifies Na/K-ATPase and catecholamine sensitivity → episodic massive K shift with acute proximal weakness, classically in young ASIAN/Hispanic MALES. Always check TSH/free T4.

BOARD TRAP

Treating only the K and missing the thyroid. The definitive fix is controlling the hyperthyroidism (and a non-selective beta-blocker like propranolol aborts attacks); give only modest K because rebound hyperkalemia follows resolution.

9. Familial periodic paralysis

WHAT YOU SEE

Familial hypoK periodic paralysis figure

WHAT IT ENCODES

Hypokalemic periodic paralysis (autosomal-dominant CACNA1S/SCN4A channelopathy): episodic flaccid weakness triggered by carbohydrate-rich meals, rest after exercise, or stress, presenting in the teens/20s. Total body K normal.

BOARD TRAP

Loading these patients with large KCl doses. Like TPP it is a shift — give small amounts to abort the attack and prevent the rebound hyperkalemia that follows; acetazolamide is used for prophylaxis.

10. Propranolol / rebound caution

WHAT YOU SEE

Propranolol bottle; cautious avoid rebound hyperK, rest after stress

WHAT IT ENCODES

Non-selective beta-blockade (propranolol) blocks the catecholamine-driven Na/K-ATPase shift, aborting thyrotoxic and familial periodic-paralysis attacks. K is given cautiously alongside because the deficit is apparent, not real.

BOARD TRAP

In any shift-only state, full 'replacement-dose' KCl overshoots once K redistributes back out → rebound hyperkalemia. The principle: small, monitored K doses for shift; full replacement only for true total-body deficits.

11. Barium / toxin shift

WHAT YOU SEE

Barium poison skull; barium toxicity blocks K leak channels, high carb meal stress

WHAT IT ENCODES

Soluble barium salt toxicity blocks the passive K efflux (leak) channels while the Na/K-ATPase keeps pumping K in → K trapped intracellularly → acute severe hypokalemia with paralysis. Cesium and toluene are related toxic causes.

BOARD TRAP

Attributing barium-poisoning weakness to a primary neuromuscular disease. The clue is rapid profound hypokalemia after ingestion/occupational exposure; treat with K replacement and supportive care as the toxin clears.

12. Anabolic states / new cells

WHAT YOU SEE

Anabolic states, new cells eat K; B12/GM-CSF treatment

WHAT IT ENCODES

Rapid cell production consumes K as new cells form. Refeeding syndrome, B12/folate treatment of megaloblastic anemia, and GM-CSF therapy all trigger brisk intracellular uptake of K (along with phosphate and magnesium) as proliferating cells incorporate it.

BOARD TRAP

Refeeding syndrome: when feeding restarts, insulin surges and proliferating cells pull in K, phosphate, and magnesium simultaneously — anticipate and pre-replete all three, monitoring closely, rather than reacting after the crash.

13. Decreased intake

WHAT YOU SEE

Top floor: decreased intake, starvation, clay ingestion / pica

WHAT IT ENCODES

Low intake (starvation, alcoholism, clay/geophagia pica) alone rarely causes hypokalemia because the kidney can drop urine K to <15 mmol/day. It mainly compounds a coexisting shift or loss rather than being the sole driver.

BOARD TRAP

Blaming diet for marked hypokalemia. Pure dietary deficiency is a weak sole cause — if K is significantly low, look for a hidden shift (insulin, beta-agonist) or a loss (diuretic, vomiting) rather than stopping at 'poor intake.'